



Fig. 3: Author's prototype wired on Vero board



Fig. 4: Installation close to the railway track

PARTS LIST

Semiconductors:

Board1 - Arduino Nano

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 4.7-kilo-ohm

Miscellaneous:

- ANT.1 - Metal strip antenna
- DS18B20 temperature sensor
- A6 GSM module
- 5V, 2A power supply

Vin and ground pins to power it from external power supply.

For programming the source code using Arduino IDE, first connect the Arduino board to a computer through USB and then select Arduino Nano board and COM port from Tools menu. Then upload the code (labtest1rail-track.ino) to Arduino board.

Note that first you need to include the *SoftwareSerial.h*, *OneWire.h*, and *Dallas Temperature.h* in Arduino IDE before compiling and uploading the source code. You also need to include two mobile numbers in the source code: one mobile number for sending the alert messages and another mobile number for receiving the messages.

Construction and testing

After assembling the circuit on a Vero board, connect it to a 5-volt DC power supply rated at two amperes—like a mobile charger. Now, as per the program, the set up condition will start and send AT code ATH to stop any incoming call (in SIM card), so that it may enter into the loop section.

In loop section, it will again send an ATH code and then send AT command to the GSM module for sending a text message. After the sending process is complete, it includes the temperature sensor data, packs it, and then sends alert message, such as “Rail Track Temperature is 40°C.”

Note that you need to connect a metal strip as antenna to the GSM module. The author's prototype with antenna (ANT) assembled on a Vero board is shown in Fig. 3. Enclose the circuit in a suitable plastic box and install it close to the railway track to be monitored, as shown in Fig. 4. **EFY**

connect to the mobile network. The SIM card in GSM module should get uninterrupted SMS service.

Software and programming

Arduino Nano is a compact board based on ATmega328 microcontroller that has easy USB connectivity for programming on Arduino IDE. It has

EFY Note

The source code of this project is available for free download at source.efymag.com



Kamal Das is a track maintainer in Eastern Railway and also an electronics hobbyist